Request for Freight Train Operating Data

Validating a Longitudinal Train Dynamics Model for Fuel-Efficient Train Control

George James
School of Engineering, Colorado State University Pueblo

September 30, 2026

Summary

I am a graduate student building a fast, learned model of how freight trains move: their speed, the forces between cars, and the energy used. The model is meant to support fuel-efficient driving advice and planning. It is trained on a physics-based simulator, and the most important open question in my work is how closely that simulator matches real trains.

I am asking your organization for a modest amount of recorded operating data (locomotive event-recorder downloads and the matching train makeup) from a handful of ordinary freight trips. I would use it only to check and calibrate the model. I will handle it under whatever confidentiality terms you require, and I will share the results with you. Section lists exactly which data would help, in priority order. Even the first tier alone would be valuable.

The research

My master's thesis, *A Graph Neural Surrogate for Longitudinal Train Dynamics: Toward Fuel-Optimal Control of Freight Consists*, has three parts.

1. **A reference simulator.** A longitudinal train dynamics model that treats every car as its own mass, joined by couplers with slack and draft gear. It includes traction and dynamic braking, rolling and curving resistance, and grade, on real route profiles built from public elevation and track data. It runs on a GPU, so thousands of trips can be simulated.
2. **A learned surrogate.** A graph neural network that represents the train as a chain of cars and learns to reproduce the simulator. It is far faster than the simulator and handles any train length.
3. **A control demonstration.** Using the surrogate to plan throttle and dynamic brake settings that reduce fuel use while respecting speed limits and coupler-force limits.

Why real data matters

Nearly every result in the thesis comes from the simulator. I have verified that the code solves its equations correctly, and I am comparing it against an international benchmark of published train dynamics simulators. Neither shows that the equations describe a real train with real parameters. Measured data is the only way to establish that, and it is what would let the fuel and coupler-force results be stated with confidence rather than as results "within the model."

What data would help

The tiers are in priority order. Tier 1 by itself is enough to test the model against reality. Anything partial is still useful, and I can work with whatever format your systems export.

Tier 1: one trip, recorded

For each trip, the locomotive event-recorder download (lead locomotive, and remote units if the train runs distributed power) together with its consist list.

Parameter	Units	What it is used for
Time stamp	s	Aligning every channel
Speed	mph or km/h	The main quantity the model is checked against
Position (GPS or milepost)	lat/lon or mi	Locating the train on the grade and curve profile
Throttle notch	0–8	The driver's input; the model is run on exactly these commands
Dynamic brake setting	notch or %	Same
Brake pipe pressure	psi	When and how hard the air brakes were applied
Brake cylinder pressure, if recorded	psi	Brake force actually developed
Independent (locomotive) brake	psi	Separating locomotive and train braking
Traction motor current or tractive effort, if recorded	A or lbf	Checking the locomotive force model directly

Consist list for the same trip, front to rear:

Per vehicle	What it is used for
Position in train	The model is built car by car, in order
Locomotive model, or car type (e.g. covered hopper, tank)	Resistance, length and coupler type
Gross weight, or tare and load	Mass, the most important single parameter
Length over couplers	Spacing of masses along the train
Number of axles	Axle load, which enters rolling resistance
Distributed-power unit positions	Where remote locomotives sit

Sampling rate. Whatever the recorder stores. One sample per second is useful; faster is better.

How much. Five to twenty trips would be a strong data set. Variety matters more than volume: loaded and empty trains, steep and level territory, and if possible a trip or two with distributed power.

Tier 2: the route

If the trips cover territory I cannot reconstruct from public data:

- Track profile or track charts: grade (%) and curvature (degrees or radius) by milepost.
- Timetable or posted speed restrictions over the same miles.

- A milepost-to-coordinate reference, if position is recorded only by milepost.

Tier 3: any of these, if they exist

These are rarer, so I list them only in case they are on hand.

- **End-of-train device data**: rear brake pipe pressure and motion, which show how a brake application travels along the train.
- **Fuel or energy**: fuel consumed per trip, or fuel rate from the locomotive. This is the quantity the control chapter tries to reduce.
- **Instrumented-car measurements**: coupler or drawbar forces from a load-cell car, at any position in a train.
- **Component data**: draft gear test curves, tractive-effort and dynamic-brake curves for your locomotive models, or car net braking ratios.
- **Resistance tests**: coast-down or other train resistance measurements.

How the data would be used

- **Validation.** Run the simulator on the recorded notch and brake commands over the recorded route, and compare its predicted speed with the measured speed along the whole trip.
- **Calibration.** Estimate the parameters that public sources give only roughly: train resistance, braking force, and locomotive force at each notch. Filtering methods (Kalman-type estimators) give these with statistical uncertainty, and test whether the model's errors behave like noise or show a systematic bias.
- **Reporting.** The thesis would report aggregate results only: error statistics and fitted parameters. No raw records or identifying details would be published.

Confidentiality and handling

I understand operating data is sensitive. I will:

- use the data only for this research, and not share or redistribute it;
- store it on encrypted storage that only I can access;
- accept data with train IDs, crew information, dates and car numbers removed. Locations can also be withheld if the grade and curvature profile along the trip is provided instead;
- work under a non-disclosure or data use agreement, arranged through the university if you prefer;
- give you the chance to review any passage that describes your data before the thesis is submitted, and acknowledge your organization by name or not at all, as you choose;
- delete the data at the end of the project on request.

What you would receive

- A short report on how well the model reproduced your trains, with the calibrated parameters (resistance, braking and locomotive force estimates).
- A copy of the finished thesis.
- A briefing or call to walk through the results, if useful.

Timing and contact

Data received by [date] can be included in the thesis. A single trip with its consist list is enough to start, and I am happy to begin with a small sample before anything larger is considered.

George James	School of Engineering, Colorado State University Pueblo
Email	[your email]
Phone	[your phone]
Thesis advisor	[advisor name and title]

Thank you for considering this request.